Appendix A. Systematic literature review of grain facility-location models

Victor Rafael Rezende Celestino, Artur Guerra Rosa, Andréia Elizabeth Silva Barros, Gabriela Corsano, Luis Javier Zeballos, Rodolfo Dondo, Silvia Araújo dos Reis

This SLR synthesizes research on MILP models designed to optimize facility and hub location problems for grain warehouses, silos, and agricultural facilities with similar logistics. The review follows the PRISMA protocol (Page et al., 2021).

The objective is to analyze the structural formulation of these models, including assumptions, decision variables, objective functions, and constraints, to establish a benchmark for MILP location models in agriculture and facilitate model comparison. The analysis emphasizes models addressing Latin American scenarios due to regional logistical and infrastructural challenges. It evaluates both deterministic and stochastic formulations to determine how uncertainty is integrated into these frameworks.

The primary research question is: "What are the MILP models for facility and hub location problems in grain and agricultural product logistics?" Inclusion criteria required peer-reviewed journal articles or conference papers proposing deterministic or stochastic MILP formulations for agricultural supply chain nodes, with no restriction on publication year. The search string was translated into English, Spanish, and Brazilian Portuguese. The search covered Scopus, SciELO, Semantic Scholar, and Web of Science (WoS), with syntax adjusted for each platform. Table A1 details the search strings, criteria, and dates.

Table A1

Databases, search dates, and search strings.

Databases	Search dates	Language	Search strings
Scopus, SciELO, Semantic Scholar, Web of Science	04-10-2026	English	(warehouse* OR silo* OR storage OR facilit* OR hub*) AND (grain* OR soy* OR rice OR corn OR wheat OR crop* OR "agricultural commodit*") AND (locat* OR localiz* OR siting OR site OR sites OR allocat*) AND (MILP OR MIP OR ILP OR MINLP OR "mixed integer" OR "mixed-integer" OR "integer programming" OR "mathematical programming" OR "optimization model*") AND ("Latin America*" OR "South America*" OR Brazil* OR Brasil* OR Argentina OR Mexico OR Colombia OR Chile OR Peru OR Uruguay OR Paraguay)

Spanish	(almacen* OR almacén* OR silo* OR almacenamiento OR instalacion* OR instalación* OR hub*) AND (grano* OR soja OR soya OR arroz OR maiz OR maíz OR trigo OR cosecha* OR cultiv* OR "commodit* agricola*" OR "productos agricola*") AND (localizac* OR localizació* OR ubicac* OR ubicación* OR asignac* OR asignació* OR emplazamiento) AND (MILP OR MIP OR ILP OR "entera mixta" OR "mixto-entera" OR "programacion entera" OR "programación entera" OR "lineal entera") AND ("América Latina" OR Sudamérica OR Centroamérica OR Brasil OR Argentina OR México OR Colombia OR Chile OR Perú OR Uruguay OR Paraguay)
Portuguese Brazil	(armazem OR armazéns OR silo* OR armazenagem OR armazenamento OR instalac* OR instalaç* OR hub*) AND (grao* OR grão* OR soja OR arroz OR milho OR trigo OR safra* OR cultiv* OR "commodit* agricola*") AND (localizac* OR localizaç* OR alocac* OR alocaç* OR posicionamento) AND (MILP OR MIP OR ILP OR "inteira mista" OR "misto-inteira" OR "programac* inteira" OR "programaç* inteira" OR "linear inteira") AND ("América Latina" OR "América do Sul" OR Brasil OR Argentina OR Colômbia OR Chile OR México OR Peru OR Uruguai OR Paraguai)

Fig. A1 presents the PRISMA flow diagram for new systematic reviews, which included searches of databases, registers, and other sources, mapping the selection process across the identification, screening, and inclusion phases.

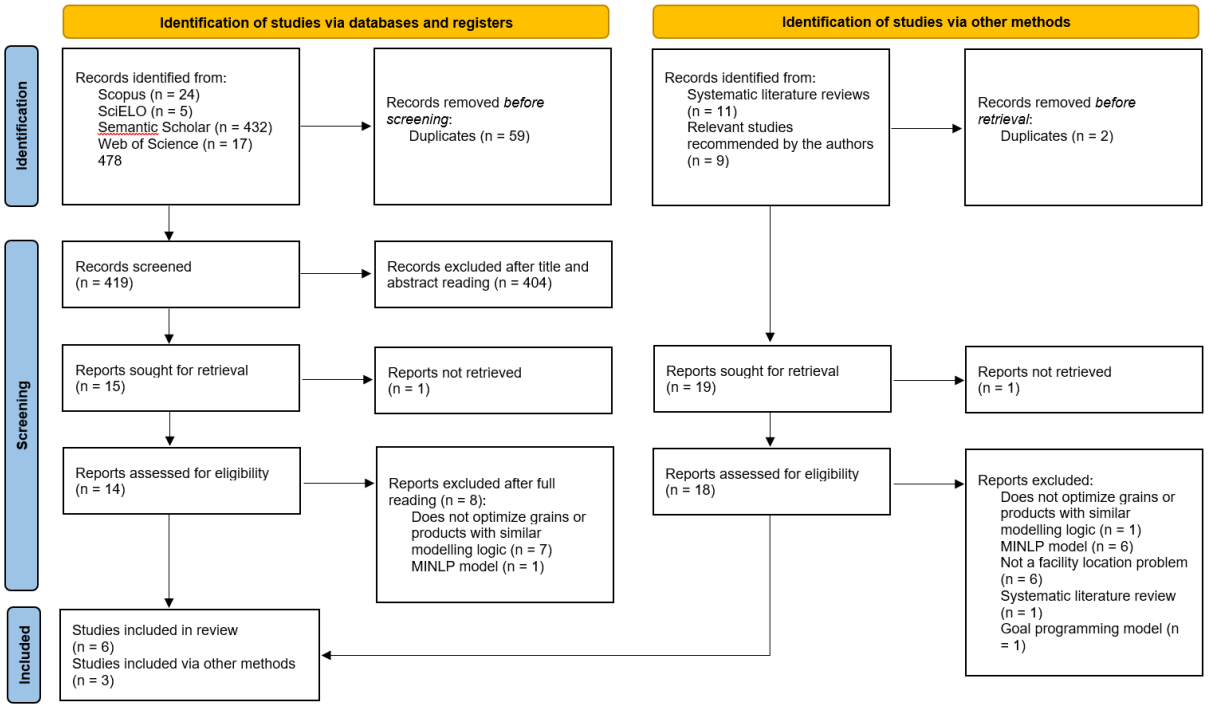


Fig. A1. PRISMA flow diagram. Source: adapted by the authors (2025) according to Page et al. (2021).

In the database search, identification yielded 478 records, including 432 from Semantic Scholar, 24 from Scopus, 17 from Web of Science, and 5 from SciELO. The removal of 59

duplicate records left 419 records for screening. Title and abstract screening resulted in the exclusion of 404 records. Of the remaining 15 reports, 1 could not be retrieved, leaving 14 for eligibility assessment. Full text evaluation excluded 8 reports because 7 did not focus on grains or products with similar modeling logic and 1 was a mixed integer nonlinear programming (MINLP) model, resulting in 6 selected studies.

For alternative sources, identification yielded 20 records, consisting of 11 systematic literature reviews and 9 author-recommended studies. The removal of 2 duplicate records left 19 reports for retrieval, of which 1 could not be retrieved. The remaining 18 reports underwent eligibility assessment, resulting in 15 exclusions: 6 were MINLP models, 6 did not address facility location problems, 1 did not optimize grains or products with similar modeling logic, 1 was a systematic literature review, and 1 used a goal programming model, resulting in 3 selected studies.

The final review includes 9 studies, comprising 6 from the database search and 3 from alternative sources.

Table A2 shows the indices and sets comparison between the models. Agricultural supply chains begin with geographic origins that vary in scale across the literature. While some frameworks define supply origins at a micro level using individual farmers or specific farm locations (Granillo-Macias et al., 2019; Vazquez et al., 2021), other models aggregate these sources into macro regions to simplify network tracking. These aggregated supply areas represent entire producing cities, regional procurement centers, or geographic centroids of grain production (Foulds, 2005; Mascarenhas et al., 2014; Mogale et al., 2020; Morais et al., 2023). Certain networks further separate these origins by the crop type grown, such as sweet and industrial (Mendoza-Ortega et al., 2021).

Table A2
Indices.

Reference	Network nodes and echelons	Time horizon	Capacity and tech levels	Transport modes and fleet	Stochastic
(Mogale et al., 2020)	Procurement centers (p), Base silos (b), Field silos (f), Regional warehouses (r), Destination warehouses (d)	Multi-period (t)	Base silo sizes (h), Field silo sizes (j)	Trucks and Rakes specific to each echelon (n_1, n_2, n_3, n_4)	-
(Milanez et al., 2026)	Producing areas (i), Distribution hubs/warehouses and Transshipment hubs (k, l, m)	Multi-period (t)*	The level of capacity of each hub (w) was not defined	Roads, rails, and rivers	-

Reference	Network nodes and echelons	Time horizon	Capacity and tech levels	Transport modes and fleet	Stochastic
(Granillo-Macias et al., 2019)	Farmers (i), Hubs (j), Malt producers (k)	Single period	Storage capacity of hubs (cap_j), Malt producers direct receiving capacity ($capm_k$)	-	Harvest yield, loss risk, penalties/incentives, hub capacities, malt producer demands (Simulated)
(Morais et al., 2023)	Soybean-production cities in Mato Grosso (i), candidate terminal locations (j), terminal types/capacity levels (s).	Single period		Road from cities to terminal; terminals prioritized by railway and/or waterway potencial.	-
(Foulds, 2005)	Supply Nodes (R_i), Intermediate Nodes (L_k), Demand Nodes (D_j)	Single period	Silo capacity (w_k)	-	-
(Mascarenhas et al., 2014)	Producing micro-regions (i), Warehouse candidate locations (j), Export ports (u)	Single Period	Capacity levels (t) for warehouses	Road (to warehouse/direct to port) and road-rail intermodal (via terminal transshipment to port)	-
(Guimaraes et al., 2017)	Production Area (i), Logistics Integrates Centers Candidates (k), Demand/delivery zones (j)	Single Period	Logistic Integration Center capacity levels (min/max)	Roads, rail and waterway	-
(Mendoza-Ortega et al., 2021)	Sweet crop producers (i), Industrial crop producers (j), Warehouses/Facilities (q), Processors (k), Distributors (h)	Single period	Uncapacitated facilities, Producers' capacities (g_i, g_j), maximum open facilities (M)	Tractor and trucks.	Sweet crop yield (V_e) and industrial crop yield (I_e) over 16 probability-based scenarios (e)
(Vazquez et al., 2021)	I Set of irrigation districts (i), F Set of farms (f), F^i Set of farms in water irrigation district i, B Set of processing facilities (b), P Set of bio-products (p), C Set of customers (c), Ω Set of adoption rate scenarios (ω).	Single period	Capacity of processing facility b (cap_b)	-	Probability of adoption rate scenario ω: Prob(ω)

Source: Authors' own work (2026).

Intermediate logistical layers show structural variation, with most models using a single layer of facilities to act as temporary warehouses, inspection hubs, or transshipment terminals (Foulds, 2005; Granillo-Macias et al., 2019; Mascarenhas et al., 2014; Morais et al., 2023). One model built a multi-echelon storage network where grain flows sequentially through multiple facilities before reaching final distribution: procurement centers, base silos, field silos, and regional warehouses (Mogale et al., 2020). To represent infrastructure constraints, studies incorporate capacity levels or discrete physical sizes for these intermediate facilities (Mascarenhas et al., 2014; Milanez et al., 2026; Mogale et al., 2020; Morais et al., 2023).

Final destinations depend on the economic purpose of the supply chain; export-oriented models terminate logistical routes at maritime seaports or international consumer markets (Mascarenhas et al., 2014; Milanez et al., 2026), whereas domestic networks route the harvest to industrial processing plants, malt producers, local consumer warehouses or general demand zones (Foulds, 2005; Granillo-Macias et al., 2019; Guimaraes et al., 2017; Mendoza-Ortega et al., 2021; Mogale et al., 2020).

Grain commodities and crop types establish another important layer. While several location models assume a single uniform commodity flowing through the network, such as barley or soybeans (Granillo-Macias et al., 2019; Guimaraes et al., 2017; Mascarenhas et al., 2014; Milanez et al., 2026; Morais et al., 2023), a smaller group explicitly models multiple product types simultaneously. This smaller multi-product group segregates crops by end use, separating raw materials or different derived bio-products (Mendoza-Ortega et al., 2021; Vazquez et al., 2021).

Time periods and uncertainty metrics are also important dimensions to these networks. Dynamic models duplicate static spatial networks across consecutive time periods to form a long-term planning horizon (Foulds, 2005), while stochastic models introduce probability-based scenarios to adjust for variations in harvest yields, weather risks, and farmer adoption rates (Granillo-Macias et al., 2019; Mendoza-Ortega et al., 2021; Vazquez et al., 2021).

Table 3 shows the parameters comparison between the models. Most models share a common core of economic and logistics parameters, such as transport costs, supply, demand, distances, and facility capacities (Foulds, 2005; Granillo-Macias et al., 2019; Guimaraes et al., 2017; Mascarenhas et al., 2014; Mendoza-Ortega et al., 2021; Milanez et al., 2026; Mogale et al., 2020; Morais et al., 2023; Vazquez et al., 2021). Only a few models go further. These add uncertainty, risk, environmental, social, or loss parameters (Granillo-Macias et al., 2019; Mendoza-Ortega et al., 2021; Mogale et al., 2020; Morais et al., 2023; Vazquez et al., 2021).

Table A3
Parameters.

Reference	Cost parameters	Logistics and capacities	Uncertainty and risk	Environmental and Social	Wastage and losses	Data Sources
(Mogale et al., 2020)	Fixed establishment (F), Transport (tc), Inventory (ih), Emission unit cost (ce), Risk penalty (rc), Loss cost (lc)	Supply (A), Demand (L), Facility capacities (cb, cf, cr), Vehicle limits (nv) and capacities (cv), Distances (dis)	Location and Transport risk factors (ϵ, g)	Carbon emission rates per vehicle/distance (e)	Transit loss fractions (lb, lj), Storage loss fractions (ls, lw)	Public reports and private field data
(Milanez et al., 2026)	Transport (C_{ij}^{p}) Open Hub (Fkw) Storage (γkp)	available flow (P_i^j), flow demanded (D_j^p), size of a hub (Γ_{kw}), collection flow (X_{ik}^p), flow traversing arcs at hub (Y_{kl}^{pm}), flow from hub (Z_{lj}^p), flow storage at hub (ε_k^p),	-	-	-	IBGE, USDA, Salin (2020), SIFRECA, CONAB, Vieira & Dalchiavon (2018), IMEA, FAO
(Granillo-Macias et al., 2019)	Freight cost (Cij, Cik, Cjk), Hub opening cost (fj), Uncertainty cost (Sn)	Available offer (sumi), Storage capacity of hub (capj), Demand of malt producer (demk), Malt producer direct capacity (capmk)	Simulation uncertainty values (Sn, Qmin), Historical yield/risk distributions	-	-	SIAP (2017), historical data (2003-2015), Standard NMX-FF-043-SCFI-2003
(Morais et al., 2023)	Terminal implementation cost (cs), cost component weight (α), existing terminal discount (Tejs)	Distance matrix (Distij), maximum road distance (Dmax), soybean demand by city (PDemi), total export demand (DemT), terminal capacity (Caps), occupation rate (ρ), average permanence time (TMP), darvest durations (Tsaf)	Not modeled.	Environmental weight (θ), waterway preference weight (Hidj) and objective weight (β), railway preference weight (Ferj) and objective weight (γ), road emission factor of 77.6 g/t·km	-	IBGE/SIDRA (2021), Bing/Google Maps (2021), logistics company data, SisDEA, PNLT (2021), Plaza et al. (2020).
(Foulds, 2005)	capacity of silo (vk), silo construction (fk),	Flows (xikt, xijt, xikt), Supply (ai, bj),	-	-	-	-

Reference	Cost parameters	Logistics and capacities	Uncertainty and risk	Environmental and Social	Wastage and losses	Data Sources
	all transportation and construction (z)	Capacity of the silo (wk)				
(Mascarenhas et al., 2014)	Road freight (c_{ij}, p_{iu}), Rail freight (a_{ju}), Port fee (CP_u), Warehouse operational cost (CO_t, H_j)	Export production capacity (M_i), Static storage capacity (cap_t), Port demand (e_u)				IBGE (2012), SIFRECA (2012), Ferrari (2006), CONAB (2013), MDIC (2014), Oliveira et al. (2012) PNLT/MT (2007, 2012), BNDES (2011), ANTT (2012), USP (2012), ANTAQ (2013), CONAB (2013), Costa (2014), EMBRAPA (2017)
(Guimaraes et al., 2017)	Direct transport cost (C^{dir}_{pijm}), inbound transport cost to LIC (C^{ent}_{pikm}), outbound transport cost from LIC (C^{lea}_{pkjm}), transshipment/modal-shift cost (CT^{lea}_{pkjm}), LIC fixed processing cost (g_{pk}), LIC variable processing cost (a_{pk})	Demand volume (d_{pij}), available transport modes $(M^{dir}_{pij}, M^{ent}_{pik}, M^{lea}_{pkj}$, LIC candidate set (K), products (P), production zones (I), demand zones (J), LIC capacities (v^{min}_k, v^{max}_k)	-	-	-	
(Mendoza-Ortega et al., 2021)	Production cost (P), Transport costs $(CT1_{iqh}, CT2_{jqh})$, Facility fixed opening cost (CFA_q), Vehicle preparation cost (CV)	Sweet crop producer capacity (g_i), Industrial crop producer capacity (g_j), Distributor demand (D_h), Processor demand (D'_k), Max facilities to open (M), Vehicle capacity (30 tons)	Scenario probabilities (φ_e), Crop yields (V_e, I_e)			
(Vazquez et al., 2021)	Capital construction cost (C^{cap}_b), Harvesting/hauling cost (C^H), Loading/unloading cost (C^L), Processing cost	Minimum and maximum facility capacity (Min_b/Max_b), Number of facilities (K), Water availability	Adoption rate probabilities	Emission factors $(\mu_b, \beta, \gamma_p, \delta_p)$, Emission price (ϵ), Emissions upper-bound (UB^{EnvImp}), Work-hour earnings	-	USDA (2018), NASS (2019), UArizona LGP (2020), ArcGIS

Reference	Cost parameters	Logistics and capacities	Uncertainty and risk	Environmental and Social	Wastage and losses	Data Sources
	(C^P), Transportation cost (C^T), Distribution cost (C_p^D)	(W_i), Irrigation use (U_f), Demand (D_{pc}), Farm-facility distance (L_{fb})		($E^B/E^H/E^T/E^P/E^D$), Work hours ($H^B/H^H/H^P/H^T/H^D$), Social impact lower-bound (LB^{SocImp})		Pro, EBID (2018), Maricopa (2019), Data USA (2019), BLS (2018), EPA (2017)

Source: Authors' own work (2026).

Economic parameters appear in all reviewed formulations, but their scope changes with the network purpose. The common core is transport cost, which links origins, facilities, transshipment points, and final demand. Most models also add fixed opening, construction, implementation, storage, processing, port, inventory, or vehicle preparation costs (Foulds, 2005; Granillo-Macias et al., 2019; Guimaraes et al., 2017; Mogale et al., 2020; Morais et al., 2023). Export models place more weight on freight, port charges, hub establishment, and storage operations (Mascarenhas et al., 2014; Milanez et al., 2026). Agro-industrial models add harvesting, hauling, loading, unloading, processing, and distribution costs (Mendoza-Ortega et al., 2021; Vazquez et al., 2021).

The logistics and capacity layer is more stable. The models usually define available supply, demand, distances, facility capacities, storage capacities, vehicle capacity, and candidate node limits. Some formulations add terminal occupation, permanence time, harvest duration, water availability, irrigation use, or maximum number of facilities (Mendoza-Ortega et al., 2021; Morais et al., 2023; Vazquez et al., 2021). These parameters convert the geographic network into an operational network, because they define how much product can be produced, moved, stored, processed, and received (Mascarenhas et al., 2014; Milanez et al., 2026; Mogale et al., 2020). Models with intermodal or corridor detail use distance matrices, modal availability, terminal capacity, and route limits. Models focused on processing add minimum and maximum facility size, crop input limits, product demand, and biomass conversion requirements (Guimaraes et al., 2017; Mascarenhas et al., 2014; Milanez et al., 2026; Mogale et al., 2020; Morais et al., 2023; Vazquez et al., 2021).

Uncertainty, risk, environmental, social, and loss parameters are less uniform. Only part of the sample includes explicit stochastic or risk terms, such as simulated uncertainty values, yield and risk distributions, crop yield scenarios, or adoption-rate probabilities (Granillo-Macias et al., 2019; Mendoza-Ortega et al., 2021; Mogale et al., 2020; Vazquez et al., 2021).

Environmental parameters appear as emission costs, emission factors, modal preference weights, or upper bounds on environmental impact (Mogale et al., 2020; Morais et al., 2023; Vazquez et al., 2021). Social parameters appear only when local work hours and earnings are modeled as a benefit (Vazquez et al., 2021). Losses are also limited, appearing through transit and storage loss fractions in one green food supply chain formulation (Mogale et al., 2020). Data sources follow the empirical setting of each model, with public agricultural databases, transport cost systems, mapping platforms, logistics reports, government agencies, and private field data used to parameterize production, distances, capacities, costs, and demand (Granillo-Macias et al., 2019; Mendoza-Ortega et al., 2021; Mogale et al., 2020; Morais et al., 2023; Vazquez et al., 2021).

Table 4 shows the decision variables in comparison between the models. Most models use two main types of variables. Binary variables set the location and allocation choices. Continuous variables measure the product flows, storage levels, and costs (Foulds, 2005; Granillo-Macias et al., 2019; Guimaraes et al., 2017; Mascarenhas et al., 2014; Mendoza-Ortega et al., 2021; Milanez et al., 2026; Mogale et al., 2020; Vazquez et al., 2021). Only a few models add integer variables to count discrete transport resources or define the model size (Foulds, 2005; Mogale et al., 2020).

Table A4
Decision variables

Reference	Binary	Integer	Continuous
(Mogale et al., 2020)	Silo establishment ($X \in \{0,1\}$), Node-to-node assignments ($Y \in \{0,1\}$)	Number of mixed-capacity vehicles/rakes utilized per link ($V \in \mathbb{Z}^+$)	Shipment quantities ($U \geq 0$), Inventory levels ($IN \geq 0$), Transit and Storage loss quantities ($Tl, Sl \geq 0$)
(Milanez et al., 2026)	opening of hub (H_{kw}), is equal to 1 if a hub is opening at node k with capacity level w, and is equal to 0.	-	collection flow (X_{ik}^{p}), flow traversing arcs at hub (Y_{kl}^{pm}), flow from hub (Z_{lj}^{p}), flow storage at hub ($\mathcal{E}k^{p}$)
(Granillo-Macias et al., 2019)	Hub j is open or not ($Y_j \in \{0, 1\}$), Incorporating uncertainty cost for candidate solution *n* ($Z_n \in \{0, 1\}$)	-	Tonnes to transport from farmers, hubs and malt producers (X_{ij}, X_{ik}, $X_{jk} \geq 0$)
(Morais et al., 2023)	Allocation of city i to terminal j (xij); installation of terminal type s at location j (yjs)	-	-
(Foulds, 2005)	silo constructed at L_k (y_k)	Number of regions that are sources of supply (m), number of final destinations of the grain	the number of units shipped directly from Ri to Lk in period t (x_{ikt}), the number of units shipped directly from Lk to Dj in period t (x_{kjt}), the number of units shipped from Ri to Dj in period t (x_{ijt}), the capacity of the silo at Lk (w_k), the total cost of all transportation and construction (z).

Reference	Binary	Integer	Continuous
		shipments (n), number of possible silo locations for intermediate storage (p)	
(Mascarenhas et al., 2014)	Installation of warehouse size t in micro-region j ($z_{jt} \in \{0,1\}$)		Transported quantities ($x_{ij}, y_{ju}, w_{iu} \geq 0$), Soy arriving at port (d_u), Warehouse operational cost (H_j)
(Mendoza-Ortega et al., 2021)	Facility q is open ($Y_q \in \{0,1\}$), Facility q assigned to distributor h ($Y1_{qh} \in \{0,1\}$), Facility q assigned to processor k ($Y2_{qk} \in \{0,1\}$)		Sweet crop flow through facility ($QV_{iqhke} \geq 0$), Industrial crop flow through facility ($QI_{jqhke} \geq 0$)
(Guimaraes et al., 2017)	LIC k is open or not ($Z_k \in \{0, 1\}$)	-	Direct shipment volume ($x_{pijm}^{dir} \geq 0$), inbound shipment volume to LIC ($x_{pikm}^{ent} \geq 0$), outbound shipment volume from LIC ($x_{pkjm}^{lea} \geq 0$), peer-to-peer volume through LIC ($y_{ipkj} \geq 0$)
(Vazquez et al., 2021)	processing facility b is added (x_b),	-	Capacity of processing facility b (cap_b), Commodity flow of biomass from farm f to processing facility b ($u^{(\omega)}_{fb}$), Commodity flow of bio-product p from processing facility b to customer c ($v^{(\omega)}_{pbc}$)

Source: Authors' own work (2026).

Binary variables form the main strategic layer of mathematical formulations. They dictate whether specific structural facilities are physically opened, installed, added, or assigned. These permanent facilities include grain silos, intermediate warehouses, export terminals, and logistics integration centers (Foulds, 2005; Guimaraes et al., 2017; Mascarenhas et al., 2014; Mogale et al., 2020; Morais et al., 2023). Other frameworks utilize binary choices to establish regional logistics hubs and raw material processing plants (Granillo-Macias et al., 2019; Mendoza-Ortega et al., 2021; Milanez et al., 2026; Vazquez et al., 2021). Some models also use binary decisions to connect product origins directly to terminals and activate exact facility types (Morais et al., 2023). Specific structures apply discrete choices to select warehouse sizes or assign intermediate facilities to final distributors and processors (Mascarenhas et al., 2014; Mendoza-Ortega et al., 2021). Advanced formulations include specific uncertainty scenarios within the candidate solutions through binary parameters (Granillo-Macias et al., 2019).

Continuous variables represent operational behavior after strategic facility decisions are made. These metrics measure exact shipment quantities across the supply chain (Foulds, 2005; Granillo-Macias et al., 2019; Mascarenhas et al., 2014; Mogale et al., 2020). Specific

formulations deploy continuous variables to track direct product flows, inbound and outbound transfers, and peer to peer movements (Guimaraes et al., 2017; Milanez et al., 2026). Other models utilize this continuous layer to monitor distinct biomass volumes and bio-product flows (Vazquez et al., 2021). These metrics also quantify inventory levels, permanent storage capacities, and precise vessel arrivals at export ports (Mascarenhas et al., 2014; Milanez et al., 2026; Mogale et al., 2020). Advanced frameworks apply continuous numbers to define structural facility capacities, calculate total network costs, and quantify exact loss quantities (Foulds, 2005; Mascarenhas et al., 2014; Mogale et al., 2020; Vazquez et al., 2021).

Integer variables are used much less often. Their practical role is mainly to count transport resources, such as the number of vehicles or train rakes used on each link (Mogale et al., 2020). Other numerical counts in the reviewed tables define the model size, such as the number of supply regions, final destinations, or candidate silo locations, rather than decisions selected by the solver (Foulds, 2005).

Table A5 shows the objective functions comparison between the models. Most reviewed models use single-objective optimization to minimize network expenses (Foulds, 2005; Granillo-Macias et al., 2019; Guimaraes et al., 2017; Mascarenhas et al., 2014; Mendoza-Ortega et al., 2021; Milanez et al., 2026; Mogale et al., 2020; Morais et al., 2023). Only one model used multi-objective formulation to minimize economic expenditures and greenhouse gas emissions while maximizing local workforce benefits (Vazquez et al., 2021).

Table A5
Objective function

Reference	Type	Economic	Risk	Environmental and social
(Mogale et al., 2020)	Single objective. Minimize.	Facility establishment (X), Transportation (U), Inventory holding (IN), Cost of transit and storage food loss (Tl, Sl)	Risk penalty costs for establishment (X) and transport (Y)	Carbon CO_2 emission costs (V)
(Milanez et al., 2026)	Single objective. Minimize.	costs of collection (CC), transfer between hubs (TRC), distribution (DC), holding (HC), additional (AC) and hub establishment (EC)	-	-
(Granillo-Macias et al., 2019)	Single objective. Minimize.	Transportation cost (X_{ij}, X_{ik}, X_{jk}), Opening cost of hubs (Y_j)	Impact on uncertainty obtained via simulation ($S_n \cdot Z_n$)	-
(Morais et al., 2023)	Single objective. Minimize.	Terminal implementarion cost ($\alpha \cdot$ UT(cost) = $\alpha \cdot [\sum j \in J \sum s \in S$ (yjs $\cdot$ cs − TEjs $\cdot$ cs)])	-	Waterway and/or railway preference and road-distane as CO_2 proxy

Reference	Type	Economic	Risk	Environmental and social
(Foulds, 2005)	Single objective. Minimize	fixed cost of silo (f_k), total cost of all transportation and construction (z), the per unit transportation cost from Lk to Dj (c_{kj}), the per unit transportation cost from Ri to Lk (c_{ik}), the per unit transportation cost from Ri to Dj (c_{ij}).	-	-
(Mascarenhas et al., 2014)	Single objective. Minimize.	Transportation costs (road and rail), port fees, and fixed storage operational costs ($\min Z = \sum c_{ij}x_{ij} + \sum a_{ju}y_{ju} + \sum p_{iu}w_{iu} + \sum CP_u d_u + \sum H_j$)	-	-
(Mendoza-Ortega et al., 2021)	Single objective. Minimize.	Production, transportation, vehicle preparation, and warehouse opening costs ($\min z = \sum \frac{P}{V_e} QV_{iqhke}\varphi_e + \sum \frac{P}{I_e} QI_{jqhke}\varphi_e + \sum C\,T1_{iqh} QV_{iqhke}\varphi_e + \sum C\,T2_{jqh} QI_{jqhke}\varphi_e + \sum \frac{QV_{iqhke}}{30} CV\varphi_e + \sum \frac{QI_{jqhke}}{30} CV\varphi_e + \sum C\,FA_q Y_q$)		
(Guimaraes et al., 2017)	Single objective. Minimize.	Direct transportation cost (x^{dir}), inbound transportation cost to LIC (x^{ent}), outbound transportation cost from LIC (x^{lea}), transshipment/modal-shift cost (CT^{lea}), fixed and variable LIC processing costs (z_k, x^{ent})	-	-
(Vazquez et al., 2021)	Multi-objective. Minimize and maximize.	Capital cost and expected operational cost over adoption-rate scenarios ($EcoBen$)		Construction and expected operational GHG emissions ($EnvImp$); local job creation from construction and operational work hours and earnings ($SocImp$)

Source: Authors' own work (2026).

Economic factors are mainly addressed by minimizing transportation and distribution costs between farms, intermediate hubs, and final destinations (Foulds, 2005; Granillo-Macias et al., 2019; Mascarenhas et al., 2014; Mendoza-Ortega et al., 2021; Milanez et al., 2026; Mogale et al., 2020). The formulations include fixed and variable capital requirements for infrastructure, such as silo construction based on storage capacity or intermodal transshipment terminal implementation (Foulds, 2005; Morais et al., 2023). Operational expenses incorporate port tariffs for maritime exports, vehicle preparation charges, processing expenditures, and inventory holding fees (Mascarenhas et al., 2014; Mendoza-Ortega et al., 2021; Milanez et al.,

2026; Vazquez et al., 2021). One work also account for the financial loss from food spoilage during transit and storage (Mogale et al., 2020).

Risk integration is less common in the objective functions. Penalty costs can be associated with facility establishment and transportation risks to account for logistical hazards (Mogale et al., 2020). Risk can be indirectly handled by evaluating the financial impact of operational uncertainty through hybrid simulation outputs, testing variables like stochastic harvest yields, weather loss risks, and delivery quality penalties (Granillo-Macias et al., 2019). The remaining models use deterministic variables and parameters in the objective function, without quantifying risk or disruption elements.

Environmental and social impacts also appear in few optimization models. Environmental objectives target the reduction of carbon dioxide and greenhouse gas emissions generated during facility construction and logistical operations (Mogale et al., 2020; Vazquez et al., 2021). The use of shorter road distances and a preference for waterways or railways are observed as proxies for mitigating environmental damage (Morais et al., 2023). Social elements, though rarely included, can be modeled as an objective to maximize local job creation by quantifying the estimated work hours and regional earnings required for processing facility construction and agricultural operations (Vazquez et al., 2021).

Table A6 shows the constraints comparison between the models. Models maintain supply chain continuity by equating incoming and outgoing delivery volumes (Foulds, 2005; Granillo-Macias et al., 2019; Mascarenhas et al., 2014). Intermediate accumulation is prevented by mandating zero stock levels at temporary warehouses (Mascarenhas et al., 2014). The volume of inputs leaving farms must match the volume of processed goods reaching final customers (Vazquez et al., 2021). Time periods are linked by adding previous inventory to new arrivals and subtracting outbound shipments to determine current stock (Milanez et al., 2026; Mogale et al., 2020). This inventory tracking includes deductions for product degradation over time (Mogale et al., 2020). Simplified network designs calculate total export requirements without enforcing flow conservation across intermediate layers (Morais et al., 2023).

Table A6
Constraints

Reference	Mass balance and flow conservation	Capacity and demand limits	Logical and assignment rules	Unique
(Mogale et al., 2020)	Inventory balance across periods ($IN^{t-1} + \sum U_{in} - \sum U_{out} - Sl = IN^{t}$)	Supply limit ($\sum U \leq A$), Facility capacity ($IN \leq c \cdot X$), Vehicle capacity ($U \leq cv \cdot V$), Fleet availability	Big-M flow bounds ($U \leq Z \cdot Y$), Assignment requires establishment ($Y \leq X$), Linking logic	Transit loss calculation ($Tl = l \cdot U$), Storage loss

Reference	Mass balance and flow conservation	Capacity and demand limits	Logical and assignment rules	Unique
		($\sum V \leq nv$), Exact demand satisfaction ($\sum U = L$)	($\sum X \geq, \sum X - M \leq$), Max one size per location ($\sum X \leq 1$)	calculation ($Sl = l \cdot IN$)
(Milanez et al., 2026)	storage level at the end of the previous period (p-1) plus all the incoming flow minus the flow that leaves the hub	Demand Limits: guarantees that all the demand in period p is attended by a hub	Flow sent from a port to a final consumer (Z_{lj}^{p}) can only exist if that port is an open hub (H_{lw} =1).	
(Granillo-Macias et al., 2019)	Hub inflow/outflow balance ($\sum X_{ij} = \sum X_{jk}$)	Demand satisfaction ($\sum X_{jk} + \sum X_{ik} \geq dem_k$), Supply limit ($\sum X_{ij} + \sum X_{ik} \leq sum_i$), Hub storage capacity ($\sum X_{ij} \leq cap_j$), Malt producer direct capacity ($\sum X_{ik} \leq capm_k$)	Logic constraint mapping hub opening to capacities ($\sum X_{jk} \leq Y_j \cdot cap_j$), Incorporating uncertainty logic for iterations (Z_n)	Objective ceiling limit based on optimal simulated uncertainty ($Z \leq Q_{min}$
(Morais et al., 2023)	Total export demand (∑i∑j xij·PDemi = DemT), no multi-echelon flow conservation.	Distance cap (xij·Distij ≤ Dmax) and terminal capacity constraint (∑s Caps·yjs·ρ ≥ ∑i xij·PDemi·TMP/Tsaf).	Terminal type per location (∑s yjs ≤ 1), one terminal per city (∑j xij ≤ 1), assignment only to installed terminals (xij ≤ ∑s yjs), self-assignment for installed terminals (xjj = ∑s yjs), binary variables.	Georeferenced road-distance constraint; terminal capacity adjusted by occupancy rate, permanence time, and harvest duration.
(Foulds, 2005)	Silo material flow balance ($\sum x_{ikt} = \sum x_{kjt}$)	Supply limit ($\sum x_{ikt} = a_i$), demand requirement ($\sum x_{kjt} = b_j$), dynamic continuous silo throughput capacity ($W_k = \sum x_{ikt}$)	Big-M capacity linkages ($My_k \geq W_k$), domain requirements ($y_k \in \{0,1\}$, $x_{ikt}, x_{kjt}, W_k \geq 0$)	
(Mascarenhas et al., 2014)	Flow conservation / zero stock at warehouses ($\sum x_{ij} = \sum y_{ju}$)	Supply limit ($\sum x_{ij} + \sum w_{iu} \leq M_i$), Warehouse dynamic capacity ($\sum x_{ij} \leq \sum cap_t z_{jt}$), Port demand ($\sum y_{ju} + \sum w_{iu} \geq e_u$)	Warehouse operational cost formulation ($H_j = \sum CO_t z_{jt}$), Domain limits ($z_{jt} \in \{0,1\}$, $x_{ij}, y_{ju}, w_{iu} \geq 0$)	-
(Mendoza-Ortega et al., 2021)		Sweet crop yield capacity ($\sum Q\, V_{iqhke} \leq V_e \cdot g_i$), Industrial crop yield capacity ($\sum Q\, I_{jqhke} \leq I_e \cdot g_j$), Sweet crop demand assignment ($\sum Q\, V_{iqhke} \leq D_h \cdot Y1_{qh}$), Industrial crop demand assignment ($\sum Q\, I_{jqhke} \leq D'_k \cdot Y2_{qk}$)	Max open facilities limits ($\sum Y\,1_{qh} \leq M, \sum Y\,2_{qk} \leq M$), Assignment linkages ($Y1_{qh} = Y_q, Y2_{qk} = Y_q$), Domain rules ($QV_{iqhke}, QI_{jqhke} \geq 0, Y_q, Y1_{qh}, Y2_{qk} \in \{0,1\}$)	

Reference	Mass balance and flow conservation	Capacity and demand limits	Logical and assignment rules	Unique
(Guimaraes et al., 2017)	Demand balance: direct shipments plus LIC-mediated shipments must satisfy each O/D demand $\left(\sum x_{pijm}^{dir} + \sum y_{ipkj} = d_{pij}\right)$. LIC flow conservation links inbound flows to peer-to-peer LIC flows and outbound LIC flows $(\sum y = \sum x^{ent}; \sum y = \sum x^{lea})$	LIC capacity bounds activated by opening decision $\left(v_k^{min} z_k \leq \sum x_{pikm}^{ent} \leq v_k^{max} z_k\right)$; exact satisfaction of O/D demand $\left(d_{pij}\right)$	Non-negativity of flow variables $(x^{dir}, x^{ent}, x^{lea}, y \geq 0)$; binary LIC opening $(z_k \in \{0,1\})$; flows through LIC only possible when $z_k = 1$due to capacity linking	Strict peer-to-peer demand enforcement: the model preserves each origin-destination pair through y_{ipkj}, preventing artificial mixing of flows at LICs even if mixing would reduce costs
(Vazquez et al., 2021)	the biomass commodity flow from the farms f to the processing facilities matches bio-products p commodity flow from the processing facility b to the customers c;	To design the guayule and guar supply chains, certain limitations and requirements need to be included. The proposed model considers planning, harvesting, processing, demand, transportation, and distribution, requirements as well as environmental and social limitations. The maximum number Kof processing facilities to be added is ensured by Σ b∈B x b ≤K; and the processing facilities' capacity cap b bounds by $Min_b x_b \leq cap_b \leq Max_b x_b$, ∀b; the farming capacity and the water availability as presented by Σ b∈B $u^{(\omega)}_{fb} \leq P^{(\omega)}_f$, ∀f, ∀ω and Σ f∈F i Σ b∈B $u^{(\omega)}_{fb}$ $U_f \leq W_i$, ∀i, ∀ω, respectively. The required distribution to satisfied the demand of each bio-product p from each customer cis determined by Σ b∈B $v^{(\omega)}_{pbc} \geq D_{pc}$, ∀p, ∀c, ∀ω	If no processing facility is added, constraints , $u^{(\omega)}_{fb} \leq Max_b x_b$, ∀f, ∀b, ∀ω and Σ p∈P Σ c∈C $v^{(\omega)}_{pbc} \leq Max_b x_b$, ∀b, ∀ω ensure that transportation and distribution are not available, respectively. the model's design requirements consider binary xb ∈ {1, 0}, ∀b and nonnegativity cap_b, $u^{(\omega)}_{fb}$, $v^{(\omega)}_{pbc} \geq 0$, ∀f, ∀b, ∀p, ∀c limitations for the decision variables.	A maximum dimension of the environmental impact EnvImp is ensured by EnvImp ≤ UB^{EnvImp} through an upper-bound for the emissions cost. Similarly, a minimum social impact SocImpis considered in SocImp ≥ LB^{SocImp} by establishing a lower-bound to the total social benefit obtained.

Source: Authors' own work (2026).

Physical constraints define the operational boundaries of agricultural logistics by restricting the volume of raw materials drawn from farms to avoid exceeding local harvest yields (Mascarenhas et al., 2014; Mogale et al., 2020; Vazquez et al., 2021). Processing plants and storage hubs feature throughput ceilings to cap simultaneous product handling (Granillo-Macias et al., 2019; Mendoza-Ortega et al., 2021; Milanez et al., 2026), with separate yield

limits established for distinct crop varieties (Mendoza-Ortega et al., 2021). Distribution constraints force the network to satisfy consumer demand targets across destination points, which makes the demand one of the main objectives that must be met (Foulds, 2005; Mogale et al., 2020; Vazquez et al., 2021). Transportation networks enforce load limits for individual vehicles alongside caps on total fleet sizes (Mendoza-Ortega et al., 2021; Mogale et al., 2020), while planners limit the travel distance permitted between origins and destinations (Morais et al., 2023).

Binary variables determine facility installation and operational status within the supply chain network. Models link these opening decisions to operational flows to prevent shipments to inactive sites (Granillo-Macias et al., 2019; Milanez et al., 2026; Mogale et al., 2020) and limit the total number of processing hubs permitted within the simulated region (Mendoza-Ortega et al., 2021; Vazquez et al., 2021). Assignment logic requires farmers and distributors to connect exclusively to installed local terminals (Mendoza-Ortega et al., 2021; Morais et al., 2023). In some cases, it is shown the need to forbid the installation of multiple terminal sizes or types at the same site (Mogale et al., 2020; Morais et al., 2023).

Supply chain models also incorporate parameters representing physical loss and environmental and social constraints. Calculations determine the fraction of grain lost to degradation during transportation and silo storage (Mogale et al., 2020). Environmental policies limit greenhouse gas emissions, while social sustainability metrics establish minimum thresholds for job creation (Vazquez et al., 2021). Other constraints address specific operational scenarios: for instance, slack variables allow seaport capacities to be exceeded at a penalty cost, enabling models to represent port congestion (Milanez et al., 2026). Finally, terminal capacity evaluations adjust limits based on site occupancy rates, crop permanence time, and harvest season duration (Morais et al., 2023).

The following literature complements the SLR, which were added due to their importance in the area but were outside the SLR criteria. A literature review of harvest and production planning models indicates a reliance on deterministic approaches and the need for models that account for operational uncertainties and perishability (Taşkıner & Bilgen, 2021). Another SLR states that mathematical models for grain warehouse locations optimize logistics in developing countries, though research lacks disruption scenarios and comprehensive sustainability metrics (Rosa et al., 2025).

In facility location and network design, the locations and capacities of new grain silos are optimized to minimize transportation costs across storage-deficient regions (Steiner Neto et al., 2017). Large-scale silo location networks reduce post-harvest losses and maximize social

sustainability by measuring job creation and farmer welfare (Mogale et al., 2023). To manage uncertainty and disruption, intermodal food grain transportation networks account for unexpected hub failures, maintaining a continuous food supply by rerouting delayed shipments through designated emergency hubs (Maiyar & Thakkar, 2019). For harvest and production planning, farm-level operations connect directly to supply chain design. Integrating harvest scheduling and crop establishment into the bioenergy distribution network ensures a steady, continuous raw material supply (Anselmino et al., 2025).

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